PEDIATRIC CARE

KPPC Multisystem Inflammatory Syndrome in Children (MIS-C) Pathway for Stable, Non-Toxic Patients[1]

If the child (age 0-18) stable and non-toxic[2] and has:

Fever 38C or higher for ≥72 hours AND

Has 2 of the clinical features

MIS-C Definition

OR

Has a known close COVID contact[3] or primary illness in the last 4 weeks

OR

Has 5 or more days of fevers with no other symptoms

Differential

YES

NO

Patient should be followed up in clinic in 24-48 hours if still febrile, clinically worsening, or developing further symptoms

Obtain 1st Tier labs:

1. CBC with Diff
2. Lytes, BUN, creatinine
3. CRP and ESR
4. COVID PCR swab
5. COVID serology
6. BCX, UA, UCX

Abnormal Labs:

CRP >30 mg/L
ESR > 40 mm/hr
Na < 135 mEq/L
Plt < 150,000/mcl
ALC < 1000/mcl
ANC > 8000/mcl

NO

YES

- If the suspicion for early MIS-C necessitates further guidance for next steps, including 2nd tier testing, hospital observation, or follow up plans, contact local Pediatric ID specialist or use PTAC to contact Pediatric Hospitalist for recommendations.

- If 2nd Tier testing is recommended, would consider:

1. BCX, UA, UCX if not already collected
2. EBV serology, ASO, Bacterial GI panel if predominantly GI symptoms
3. Albumin, CK, AST, ALT, T/D bili, PTT, INR
4. Fibrinogen, triglycerides, D-dimer, LDH
5. Ferritin
6. BNP and troponin

Outpatient-contact PTAC to arrange transfer if admission indicated
Unstable, Toxic Patient-go to Inpatient, ICU and ED pathway

PTAC
562-658-3980
8-137-3980

Clinical Features for MIS-C[4]

1. GI symptoms – diarrhea, emesis, abdominal pain
2. *Rash
3. *Conjunctivitis – typically limbic sparing and non-exudative
4. *Oral mucosal changes – erythematous cracked lips, strawberry tongue, erythema of oral or pharyngeal mucosa
5. *Cervical lymphadenopathy - > 1.5cm, usually unilateral
6. *Erythema or edema of hands or feet
7. Neurologic symptoms, headaches, irritability, lethargy, AMS

[1]Patients should be seen in person in clinic for evaluation

[2]For sick/ill appearing patient, please refer to "ill" pathway

[3]Close COVID contact = <6ft for >15 minutes

[4]Please consider differential diagnosis. Note that classic and Incomplete Kawasaki Disease (KD) should strongly be considered in children presenting with 5 or more days of fever and 4-5 of the clinical criteria for classic KD, or with 2-3 clinical criteria for incomplete KD, as well as infants with 7 or more days of fever with no other explanation. *Denotes the Kawasaki Disease clinical criteria above.

Stable Pathway approved 6.30.2020, Updated 8.3.2020
Subcommittee: Dr. Paul Han, Dr. Mark Salzman, Dr. Zackary Taylor, Dr. Evan Steinberg, Dr. Judith Garza, Dr. Anusha Ramanathan, Dr. Rhina Castillo, and Dr. Joseph Mares

Unstable Pathway

PEDIATRIC CARE

INPATIENT, ICU AND EMERGENCY DEPARTMENT PATHWAY FOR MULTISYSTEM INFLAMMATORY SYNDROME (MIS-C)-UNSTABLE PATIENTS

If the child (age 0-18) ill-appearing and has:

Fever 38C or higher for ≥72 hours AND

Has 2 of the clinical features OR

Has a known close COVID contact[3] or primary illness in the last 4 weeks OR

Has 5 or more days of fevers with no other symptoms

↓

Consider differential

↓

Treat possible sepsis/ shock

-ABCDEs, Vitals, O2
-IV access-2 large proximal PIV, IO if unable to obtain PIV
-Obtain Tier 1 and 2 labs and those based on differential
-Consider POC echo if able

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Fluids (if unstable):
1. Bolus 20 ml/kg crystalloid – UNLESS concerned for cardiogenic shock (poor function on bedside focused echo, crackles, hepatomegaly, cardiomegaly on CXR)
2. If concerned for cardiogenic shock give 5-10 ml/kg bolus and IMMEDIATELY reassess (w/in 15 min) and repeat if no sign of worsening perfusion
3. Consult intensivist if not improving to initiate inotropes
4. Consider need for ECMO in refractory shock or unstable arrhythmias not responsive to standard medical therapy or pressors

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Antibiotics (after obtaining blood culture): Ceftriaxone/Clindamycin +/- Vancomycin

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Contact pediatric intensivist via PTAC for advice and to arrange transport to KPPC Tertiary Center

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Consults as indicated:
Pediatric intensivist
Infectious Disease
Cardiology
Rheumatology

Return to Stable Pathway

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Clinical Features for MIS-C

1. GI symptoms – diarrhea, emesis, abdominal pain
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7. Neurologic symptoms, headaches, irritability, lethargy, AMS

Patient Transferred

↓

Inpatient Management

↓

Obtain Studies
1. Echocardiogram-within 24 hrs, or urgently if concerns of shock*
2. Chest X-ray
3. EKG*
4. Ensure all needed 1st and 2nd Tier labs obtained

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Medications:
1. IVIG 2g/kg (to max of 100g)*
2. Solumedrol 1mg/kg/dose q12h to max of 30mg IV q12h x3-5 days

↓

Consults:
1. Rheumatology
2. Infectious Disease
3. Cardiology-if significant coronary artery involvement, LV dysfunction, arrhythmia, or if clinical myo/pericarditis
4. Neurology/Nephrology as indicated

↓

Trend labs:
1. Repeat CBC, manual differential, CRP, lytes, Albumin, D-dimer, Fibrinogen, Ferritin, AST/ALT daily.
2. Trend Troponin/BNP daily if abnormal until improved OR can do q48h until afebrile, if initial set normal
3. Can space out labs to q48h if considerably downtrending and patient afebrile with clinical improvement

↓

Anticoagulation:
1. Aspirin 3-5mg/kg daily to max of 81mg daily as long as platelets >80k
2. Lovenox *may* be recommended by hematology (eg if thrombus or high risk of thrombosis per hematology) or by cardiology.* If Lovenox recommended, start with goal anti-Xa level of 0.5-1 and continue as outpatient per heme/cardiology recommendations.

↓

Discharge pending improvement in labs x48hrs, clinical improvement and afebrile x48hrs and tolerating oral steroids*

*See Inpatient Management for further details

PEDIATRIC CARE

APPENDIX 1: DEFINITION OF MULTISYSTEM INFLAMMATORY SYNDROME IN CHILDREN (MIS-C)

What is Multisystem Inflammatory Syndrome in Children (MIS-C)?

- MIS-C is a rare, novel condition occurring in children that results in serious illness and may rapidly deteriorate.
- It is believed to result from an exaggerated post-infectious immune response, occurring a few weeks after a COVID- 19 infection.
- Diagnosis is tricky because the child may have been relatively asymptomatic with the original acute COVID-19 infection.
- MIS-C may mimic Kawasaki disease (KD), toxic shock syndrome, and septic or hypovolemic shock from GI losses. MIS-C may also mimic atypical appendicitis or febrile gastrointestinal illnesses.
- Age of presentation ranges few months old to 21 years+, average 8 years old (vs KD is typically < 5 years old).

CDC Case Definition for Multisystem Inflammatory Syndrome in Children (MIS-C)

- An individual aged <21 years presenting with fever (≥38.0°C for ≥24 hours, or report of subjective fever lasting ≥24 hours),
- Laboratory evidence of inflammation (CRP, ESR, fibrinogen, procalcitonin, d-dimer, ferritin, LDH, IL-6, elevated neutrophils, lymphopenia, low albumin)[i], and
- Evidence of clinically severe illness requiring hospitalization, with
- Multisystem (≥2) organ involvement (cardiac, renal, respiratory, hematologic, gastrointestinal, dermatologic or neurological);

AND

- No alternative plausible diagnoses;

AND

- Positive for current or recent SARS-CoV-2 infection by RT-PCR, serology, or antigen test;

OR COVID-19 exposure within the 4 weeks prior to the onset of symptoms

When to consider MIS-C:

- Fever > 38°C without another source ≥24 hours, *especially if persistent for ≥4 days.*
- Fever > 38°C with persistent abdominal pain, with or without vomiting or diarrhea.
- Fever > 38°C with shock, clinically severe illness, or multisystem (>2) organ involvement.
- Fever > 38°C with complete (or incomplete) stigmata of Kawasaki disease.
- Fever persisting ≥ 4 days AND the presence of at least 2 of the following 5 principal KD features:
  - Rash (Polymorphous exanthema)
  - Conjunctivitis (Bilateral, painless bulbar conjunctival injection without exudate)
  - Mucosal changes (Cracking of lips and/or erythema of oral cavity or strawberry tongue)
  - Cervical lymphadenopathy (≥1.5 cm in diameter)
  - Extremity changes (Erythema and/or edema of hands/feet or desquamation of fingertips)

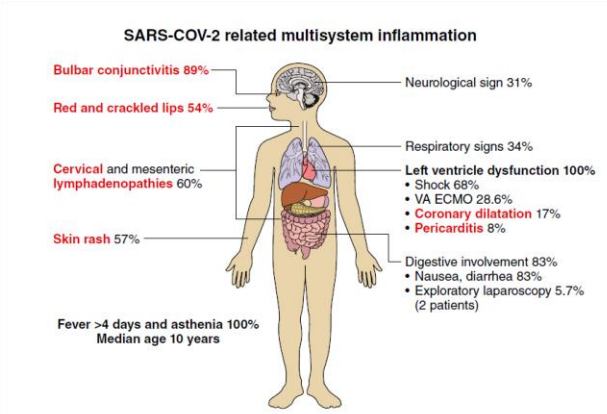

Figure Legend:

Red text are signs or symptoms consistent with Kawasaki disease or in MIS-C.

Black text, are signs that are rare in Kawasaki disease, but may be seen still in MIS-C.

(Figure from Belhadjer Z, et. al. Acute heart failure in multisystem inflammatory syndrome in children (MIS-C) in the context of global SARS-CoV-2 pandemic. *Circulation*. Published online May 17, 2020)

COVID-19:

- COVID-19 initial symptoms are fever, cough, nasal congestion or rhinorrhea, sore throat, shortness of breath, diarrhea, nausea or vomiting, fatigue, headache, myalgia, poor feeding or poor appetite

Return to Unstable Pathway

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PEDIATRIC CARE

APPENDIX 2: DIFFERENTIAL DIAGNOSIS*

DIFFERENTIAL DIAGNOSIS	CLINICAL FEATURES
Septic Shock	Can present with fevers, depending on infectious agent may have rashes, etc. This is more common than MIS-C, so should not be missed. It is important to suspect sepsis in a patient with tachycardia out of proportion to the degree of fever, and evidence of decreased peripheral perfusion. Hypotension is a late finding of septic shock in children.
	Patients appear clinically ill
	Associated with vital sign abnormalities, poor perfusion, and mental status changes
	Evaluate for focal source, e.g., skin/soft tissue, meningitis, pneumonia
	Obtain appropriate bacterial cultures
Toxic Shock Syndrome	Staph and strep toxin induced condition with
	Fever
	Vomiting/diarrhea
	Rash
	Confusion
	Conjuncitivits
	Headache
	Hypotension
Scarlet Fever	Fever
	Sore and red throat, some times with exudates, difficulty swallowing
	Enlarged glands
	Nausea, vomiting
	Headache
Steven Johnson Syndrome	Syndrome secondary to infection or reaction to some medications
	Fever and flu-like symptoms initially
	Widespread skin pain with red rash
	Blisters on skin and mucus membranes
	Red, watery eyes often with pain
Kawasaki's Disease	Commonly presents in children under the age of 6 and share many clinical features of MIS-C
	Fever with notable irritability
	Rash
	Non-purulent conjunctival injection
	Cervical lymphadenopathy
	Extremity changes including edema
Viral Infections	Nonspecific presentation of MIS-C may overlap with many viral illnesses
	Adenovirus
	Coxsackie/Enterovirus (particularly in the summer months)
	EBV
	HHV6 or HH7 (which cause Roseola) in infants/young children
Staph Scalded Skin Syndrome (SSSS)	Common in children under the age of 6
	Significant skin pain and erythema of the skin folds
	Generalized erythema of the skin as the disease progresses
	Flaccid bullae with erosions at the area of friction
	Fever, irritability and malaise
Myocarditis	Clinical signs of cardiogenic shock (cold extremities, diminished pulses/capillary refill, hypotension)
	Elevated troponin
	ST changes on EKG
	Cardiac dysfunction on echo
	Often associated with arrhythmias
Serum Sickness/Drug Reaction	Commonly presents following exposure to a drug or therapeutic agent
	Fever
	Rash
	Arthralgias
Fever and GI symptoms	Consider appendicitis, pneumonia, colitis, etc.

Return to Stable Pathway

Return to Unstable Pathway

*Modified from https://www.chop.edu/clinical-pathway/multisystem-inflammatory-syndrome-mis-c-differential-diagnosis

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APPENDIX 3: ED DISPOSITION, INITIAL MIS-C TREATMENT

Treatment if clinical syndrome and exam (+ labs if obtained) fit with MIS-C:

- Supportive care
- Begin with judicious fluid bolus (begin with 5-10ml/kg NS bolus) and assess frequently following each bolus
  - Unless the clinical suspicion is very high at the outset, many children are likely to get a 20 ml/kg bolus which is very appropriate for shock resuscitation.
  - Crackles, hepatomegaly, or cardiomegaly on CXR should prompt providers to minimize fluids.
  - With septic shock fluid bolus improves tachycardia / hypotension and children should be fluid resuscitated aggressively and require frequent clinical reassessment when administering fluid boluses.
  - However, if improvement not seen (which occurs with MIS-C) consider cardiogenic shock. Inotropes should be discussed with cardiology / ICU sooner rather than later
- Multi-disciplinary team approach (PICU, pediatric rheumatology, cardiology, ID)
- Antibiotics after cultures pending confirmation of MIS-C
  - Ceftriaxone + Clindamycin +/- Vancomycin

Disposition in the ED:

- **If clinical suspicion high for MIS-C AND Ward admission planned:**
  - Early discussion recommended with pediatric hospitalist and subspecialists as necessary by contacting PTAC
  - Video teleconsultation if needed and transfer arrangements can be made by contacting PTAC. PTAC will connect referring center with pediatric center hospitalists and/or transport physician based at LAMC, DWN, ANA, and FON (SD area use current arrangements to reach pediatric hospitalists).
  - Determination of level of transport will be made at time of consultation with transport physician/hospitalist and referring physician. Unlikely to use BLS; ALS or higher. KP Transport Teams based at Fontana and Downey.
- **Pediatric ICU admission planned:**
  - Early discussion recommended with pediatric intenstivist and subspecialists as necessary by contacting PTAC
  - If troponins are elevated, then likely needs admission to PICU
  - Video teleconsultation with the tertiary center (LAMC, DWN, FON) can be obtained by contacting PTAC who will connect referring center with pediatric intensivist when assessing these patients and arranging for pediatric transport.
  - Discussion regarding mode of transportation will be agree upon (likely CCT-RN or KPPC Transport Team)
- **If there is substantial evidence for cardiogenic shock**
  - Pediatric intensivist, cardiologist, ED, and pediatric hospitalist will collaborate to make the best decision about disposition of the patient.
  - If upon evaluation ECMO is recommended, transfer may be required outside of KP system, others may be required to go to LAMC instead of DWN or FON.

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APPENDIX 4: LAB INFORMATION

Lab evaluation:

- SARS-COV2 PCR testing may be positive or negative. SARS-COV2 IgG may be detected.
- Sepsis labs (blood culture, lactic acid, DIC panel etc.) and labs to evaluate for incomplete KD:
- CRP, ESR, CBC with differential, BNP, Albumin, ALT, urinalysis, Ferritin, and DIC panel.
- If evidence of LV dysfunction, shock or severe illness, then obtain typical cardiac labs: BNP, Troponin and a full set of electrolytes to include ionized calcium, magnesium and phosphorous.
- The order-set "COVID-19 HOSP IP SCAL" Initial Labs section has all these labs.

First Tier labs:

1. CBC with differential
2. Lytes, BUN, creatinine
3. CRP and ESR
4. COVID PCR
5. COVID Serology
6. Blood culture, urinalysis and urine culture

Second Tier labs:

1. Blood culture, urinalysis, urine culture if not already collected
2. EBV serology, ASO, Bacterial GI panel if predominantly GI symptoms
3. Albumin, CK, AST, ALT, T/D bili, PTT, INR
4. Fibrinogen, triglycerides, D-dimer, LDH
5. Ferritin
6. BNP and troponin

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APPENDIX 5: INPATIENT MANAGEMENT

Suspected MIS-C: ongoing fever, supportive labs showing inflammation, multi-system involvement.
Admit to PICU if shock, hypotension/pressors or concerns for cardiac dysfunction

MANAGEMENT WHILE HOSPITALIZED

Labs:

- Ensure that Tier 2 labs sent: BCx, UA, UCx, EBV, ASO, GI bacterial panel (if predominantly GI sxs), Albumin, CK, AST/ALT, T/D bili, PTT, INR, Fibrinogen, Ddimer, TG, LDH, Ferritin, BNP, Troponin, and quant gold for TB - please send QUIGs prior to IVIG administration
- Also ensure COVID-19 PCR and COVID-19 Serology tests have been sent.
- Repeat CBC, manual differential, CRP, lytes, Albumin, Ddimer, Fibrinogen, Ferritin, AST/ALT daily.
- Trend Troponin/BNP daily if abnormal until significantly improved. OR consider q48h until afebrile, if initial set normal.
- Can space out labs to q48h if considerably downtrending and patient afebrile with clinical improvement

Chest X-ray:

- Recommended at admission.

Echo:

- Recommended at admission.
- Echo timing typically upon admission, within 24 hours, but consider need for echo more urgently if evidence of shock or if clinical concern for myocardial dysfunction in communication with pediatric cardiologist.
- Necessary stabilization and/or treatment of the patient should not be delayed pending a formal echocardiogram.
- Echocardiographic evaluation in MIS-C is typically to evaluate ventricular/valvar function, pericardial effusion, and coronary artery dimensions with indexed Z-scores as needed.
- Consider repeat echocardiograms as inpatient if significant abnormality seen on initial study or worsening clinical status as needed.

EKG:

- 12 lead EKG on admission.
- Consider repeat EKG every 2-3 days per cardiologist/intensivist/hospitalist, especially if abnormality seen on initial EKG.
- Patients with conduction abnormality should be considered for continuous monitoring / telemetry.

Consults:

- Rheumatology
- Infectious Disease
- Cardiology if significant coronary artery involvement, LV dysfunction, arrhythmia, or if clinical myo/pericarditis.
- Neurology and Nephrology in selected cases as needed.

Medications:

- Start IVIG-2 gram/kg. NOTE: IVIG infusion is typically over 12 hours, but may need to be split over 1-2 days depending on consideration of fluid sensitivity and cardiac status. The rate of IVIG infusion should initially be low in MIS-C patients who have shock with myocardial dysfunction, then increased gradually based on tolerance. Monitor for symptoms of fluid overload (respiratory distress, heart failure). Consider continuing low rate and possible diuretic use if significant myocardial dysfunction.
- Start Solumedrol 1mg/kg/dose q12h to max of 30mg IV q12h x3-5 days followed by oral prednisone taper over 3 weeks (TBD when discharge anticipated).
- Antibiotics for at least the first 48 hours until cultures negative

Anticoagulation:

- Start Aspirin 3-5mg/kg daily to max of 81mg daily as long as platelets >80k (to be continued upon discharge)
- Lovenox *may* be recommended by hematology (eg if thrombus or high risk of thrombosis per hematology) or by cardiology (eg if LV EF <35%). If Lovenox recommended, start with goal anti-Xa level of 0.5-1 and continue as outpatient per heme/cardiology recommendations.

Follow-up:

Discharge pending improvement in labs x48hrs, clinical improvement and afebrile x48hrs and tolerating oral steroids

- Will need rheumatology telehealth follow up within 1 week of discharge with repeat labs (CBC, manual differential, CRP, lytes, Albumin, Ddimer, Fibrinogen, Ferritin, AST/ALT, troponin, BNP) ordered to be done outpatient prior to rheum follow up
- Follow-up echocardiograms scheduled at 1-2 weeks and 4-6 weeks after the date of treatment.
- Steroid taper-enough for at least 2-3 weeks
- Aspirin low-dose to be continued until the 4-6 week echocardiogram (or longer if persistent abnormality)
- Follow-up with ID specialist, Rheumatologist, or cardiology, or others as indicated
- *Code patients for now as Hx of COVID19 disease Z86.19 AND Vasculitis, Unspecified I77.6*

Return to Unstable Pathway

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APPENDIX 6: REFERENCES

- References:
- https://emergency.cdc.gov/han/2020/han00432.asp
- https://www.ahajournals.org/doi/10.1161/CIRCULATIONAHA.120.048360
- https://www.rcpch.ac.uk/sites/default/files/2020-05/COVID-19-Paediatric-multisystem-%20inflammatory%20syndrome-20200501.pdf
- Muti-System Inflammatory Syndrome in Children (MIS-C) Associated with COVID-19 [formerly Pediatric Multi-System Inflammatory Syndrome (PIMS)] v5.25.20 KP NCAL
- https://www.chop.edu/clinical-pathway/multisystem-inflammatory-syndrome-mis-c-differential-diagnosis
- CDC Clinician Outreach and Communication Activity (COCA) on Clinical Management MIS-C. July 16, 2020. Slides available at: https://emergency.cdc.gov/coca/calls/2020/callinfo_071620.asp
- Clinical Guidance for Pediatric Patients with Multisystem Inflammatory Syndrome in Children (MIS-C) Associated with SARS-CoV-2 and Hyperinflammation in COVID-19, Developed by the ACR MIS-C and COVID-19 Related Hyperinflammation Task Force, Approved June 17, 2020. Document available at: https://www.rheumatology.org/Portals/0/Files/ACR-COVID-19-Clinical-Guidance-Summary-MIS-C-Hyperinflammation.pdf